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Cambridge IGCSE Combined and Co-ordinated Sciences

B3 Biological Molecules

Condensed Version — Key Concepts and High-Probability Exam Questions

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Cambridge IGCSE Biology · The most important concepts and exam questions for your test

Must-Know Concepts

Elements and Building Blocks

Feature	Carbohydrates	Fats and oils (lipids)	Proteins
Elements	C, H, O (about twice as many H atoms as C or O atoms)	C, H, O (much less O than carbohydrates)	C, H, O, N (some also contain sulfur)
Made from	Starch and glycogen are made from many glucose molecules	1 glycerol + 3 fatty acids	Amino acids (20 different kinds)

Carbohydrates: Glucose, Starch and Glycogen

- **Glucose**, **$C_6H_{12}O_6$** (6 carbon, 12 hydrogen and 6 oxygen atoms forming a hexagon shape), is found in **plants and animals**. It is used in **respiration to release energy** and **transported in the blood**.
- **Starch** (**plants only**): many glucose molecules in a long chain that coils into a spiral; the energy store in plant cells (e.g. potato), broken down to glucose again when the plant needs it.
- **Glycogen** (**animals only**): many glucose molecules in **branched** chains (not coiled like starch); the energy store in animal cells, e.g. in the liver, which helps to keep blood glucose constant.

Fats and Oils (Lipids)

- **Fats** (solid at room temperature, e.g. butter) and **oils** (liquid, e.g. olive oil) are both **lipids**: **insoluble in water**, but they dissolve in **ethanol** (the basis of the ethanol emulsion test).
- A fat molecule is **one glycerol molecule** joined to **three fatty acid molecules**.
- Uses: making **cell membranes**; **energy stores**, with more energy per gram than carbohydrate stores (roughly twice as much); and a **heat insulating layer** of fat beneath the skin of mammals.

Proteins

- A protein is a **long chain of amino acids** in a **precise sequence**. Changing even one amino acid, or its position, gives a **different protein**, so an almost infinite number of proteins can be made.
- Functions: **enzymes** (biological catalysts; all enzymes are proteins), **antibodies** (help to protect the body against pathogens), **haemoglobin** (the red pigment that transports oxygen in mammalian blood), forming **cell membranes** in all organisms, and **keratin** (hair and fingernails).

Key Terms

Term	Definition
Carbohydrates	Substances that include sugars, starch and cellulose; they contain carbon, hydrogen and oxygen
Sugars	Carbohydrates that have relatively small molecules; they are soluble in water and they taste sweet
Glucose	A sugar that is used in respiration to release energy; formula $C_6H_{12}O_6$
Reducing sugar	A sugar, such as glucose, which turns Benedict's solution orange-red when they are heated together

Term	Definition
Starch	A carbohydrate that is used as an energy store in plant cells
Glycogen	A carbohydrate that is used as an energy store in animal cells
Lipids	Substances containing carbon, hydrogen and oxygen; they are insoluble in water and are used as energy stores in organisms; another name for fats and oils
Fats	Lipids that are solid at room temperature, e.g. butter
Oils	Lipids that are liquid at room temperature, e.g. olive oil
Protein	A substance whose molecules are made of many amino acids linked together; each different protein has a different sequence of amino acids
Amino acids	Substances with molecules containing carbon, hydrogen, oxygen and nitrogen; there are 20 different amino acids found in organisms
Antibodies	Molecules secreted by white blood cells, which bind to pathogens and help to destroy them; antibodies are proteins
Pathogens	Microorganisms, such as bacteria, that cause disease
Keratin	The protein that forms hair (and fingernails)
Iodine solution	A solution of iodine in potassium iodide; it is orange-brown, and turns blue-black when mixed with starch
Benedict's solution	A blue liquid that turns orange-red when heated with a reducing sugar
Emulsion	A liquid containing two substances that do not fully mix; one of them forms tiny droplets dispersed throughout the other

Food Tests and Key Facts

Test	What you do	Negative → positive
Iodine (starch)	Add a drop or two of iodine solution to the food on a white tile; no heating	Orange-brown → **blue-black**
Benedict's (reducing sugars)	Shake the food with a little water, add Benedict's solution, heat in a water-bath at about 80 °C	Blue → **green, yellow or orange-red**
Biuret (proteins)	Add biuret solution to the food and shake	Blue → **violet (purple)**
Ethanol emulsion (fats and oils)	Shake the food with ethanol, then pour the ethanol (not the food) into distilled water	Clear → **milky white emulsion**

- Benedict's colour shows how much reducing sugar there is: blue (none, a negative result) → green (very low) → yellow (slightly higher) → orange-red, often called brick red (very high).
- Even low concentrations turn orange-red if heated long enough, so time the colour change: **the less time it takes for the red colour to appear, the more reducing sugar there is.**
- **Key numbers**: glucose $C_6H_{12}O_6$; 1 glycerol + 3 fatty acids → 1 fat or oil; 20 different amino acids.

High-Probability Exam Questions

Question 1

Identify which elements are contained in starch. **A** carbon and hydrogen only; **B** carbon, hydrogen and oxygen; **C** carbon, hydrogen and nitrogen; **D** carbon, oxygen and nitrogen. (1 mark)

Answer:

- **B** (carbon, hydrogen and oxygen): starch is a carbohydrate, made of glucose, with no nitrogen

Question 2

Glucose, starch and glycogen are carbohydrates. **(a)** Identify which substance(s) are found in plants only, and not in animals. **(b)** Describe the structure of a starch molecule. (3 marks)

Answer:

- (a) **Starch** only (glucose is found in both plants and animals; glycogen only in animals)
- (b) Made of **many glucose molecules** linked together
- (b) In a **long chain**, which **coils into a spiral** shape

Question 3

Glucose is a reducing sugar. Describe how you could test a liquid for glucose. (3 marks)

Answer:

- Add **Benedict's solution** to a sample of the liquid in a test-tube
- **Heat** the tube in a **water-bath** (at about 80 °C) for a few minutes
- Glucose present: **blue** → **green, yellow or orange-red** (brick red); no glucose: it stays **blue**

Question 4

Animals that live in very cold places, such as bearded seals, often have especially thick layers of fat underneath the skin. Suggest why. (4 marks)

Answer:

- The fat acts as a **heat insulating layer**
- Fat is a poor conductor of heat, so it **reduces heat loss** from the body to the cold surroundings
- The fat is also an **energy store**
- When food is scarce, fat is used in **respiration** to release energy, including heat to keep warm

Question 5

Fat stores contain more energy per gram than carbohydrate stores. Suggest why birds that are about to go on long migrations build up stores of fat before they fly. (4 marks)

Answer:

- A long flight needs a lot of **energy**, released by **respiration** in the flight muscles
- The bird may not be able to **feed** for long periods, so it must **carry its fuel** with it
- Fat contains **more energy per gram** than carbohydrate (roughly twice as much), so it gives the **most energy for the least extra body mass**
- A **lighter** store takes less energy to carry, so the bird can **fly further** before it stops to feed

Question 6

Ethanol is used to test for fats and oils. Describe how you would carry out the ethanol test, and what you would see if the test gave a positive result. (4 marks)

Answer:

- Cut the food into **very small pieces** and put them into a clean test-tube with a little **ethanol**
- **Shake** the tube so that any fats or oils **dissolve in the ethanol**
- **Pour the ethanol**, without any of the food, into another clean test-tube containing **distilled water**
- Positive result: a **milky white** (cloudy white) **emulsion** forms in the water

Question 7

Which element is found in all proteins, but not in carbohydrates and fats? (1 mark)

Answer:

- **Nitrogen**: proteins are made of amino acids, which contain nitrogen (sulfur is in only some proteins)

Question 8

Correct the error in each statement about the biuret test: (i) "Add biuret solution to the food and heat gently." (ii) "If a food contains protein, biuret reagent turns from purple to blue." (2 marks)

Answer:

- (i) The error is **heating**: add biuret solution to the food and **shake to mix, without heating**
- (ii) The colours are the **wrong way round**: protein turns biuret reagent from **blue to purple (violet)**

Question 9

The table shows the results of tests on four samples of food.

Food	Iodine test	Benedict's test	Ethanol emulsion test	Biuret test
A	brown	blue	milky white	blue
B	blue-black	red	colourless	purple
C	brown	blue	colourless	blue
D	blue-black	red	milky white	purple

Identify which food contains **only** fat. (1 mark)

Answer:

- **Food A**: ethanol test positive (milky white); iodine stays brown, Benedict's and biuret stay blue

Common Mistakes to Avoid

- **Wrong building blocks**: fats are not made of amino acids; starch is made of **glucose**, not "sugar".
- **Starch or glycogen?** Starch is only in plants, glycogen only in animals; glucose is in both.
- **Glycogen or glycerol?** Glycogen is a carbohydrate made from glucose; glycerol is part of a fat.
- **Heating the wrong test**: only the Benedict's test is heated, in a water-bath, because a flame could make the liquid boil suddenly and spit out. Never heat the iodine or biuret test.
- **Swapping colours**: iodine starts **orange-brown** (not blue) and ends **blue-black**. Benedict's and biuret both start blue, but Benedict's ends **orange-red** and biuret ends **violet** (purple).
- **Vague results**: write the colour you saw, such as "stayed blue", never just "no change" or "positive".
- **Benedict's for starch**: Benedict's tests for **reducing sugars**; use the **iodine** test for starch instead.

Last-Minute Checklist

- I can list the elements in carbohydrates and fats (C, H, O) and in proteins (C, H, O, N).
- I can state that starch and glycogen are made from glucose, and say where each is found.
- I can describe the structure of a starch molecule and say how glycogen is different.
- I can state that a fat molecule is one glycerol plus three fatty acids, and give three uses of fat.
- I can state that proteins are made from amino acids, why the sequence matters, and four functions.
- I can describe all four food tests and their colours, and say which one needs heating.
- I can read a table of food test results and work out what each food contains.
- I can explain why animals in cold places, such as seals, and migrating birds store fat.